

Name: _____

Student ID: _____

Grade: _____

Question:	1	2	3	4	5	6	7	8	9	10	11	Total
Points:	1	2	3	1½	1½	7	6	3	5	3	3	36
Score:												

Calculus Grade = #points of questions (1 - 5) + 1 = _____

Linear Algebra Grade = $\frac{\text{\#points of questions (6 - 11)}}{3} + 1 = \underline{\hspace{2cm}}$

Final Grade = Calculus Grade * 0.3 + Linear Algebra Grade * 0.7

Read the instructions carefully!

- The use of a calculator, book, or notes is **not** allowed.
- Answer the questions in the spaces provided with black or blue pen. If you run out of room for an answer, continue on the back of the page, but make sure the final answer starts on the front.
- You can ask for extra paper to do your calculations, but you shall not submit it, the graded answers will be the ones on these papers, so make sure the final answers are properly explained here, starting in the blank space provided and finishing on the back if necessary.
- Answers without supporting work will not be given full credit, so make sure to explain the theoretical basis of what you do for a full score.
- If you feel like you are short on time, prioritize questions you know or those with higher score count.

1. **(1 pts.) Calculus:** Evaluate the following limit:

$$\lim_{x \rightarrow \infty} \frac{x^3 + 3x^2 + 2}{2x^4 + 5x^3}$$

2. **(2 pts.) Calculus:** Show that $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$.
Hint: You can use the Squeeze Theorem.

3. **(3 pts.) Calculus:** Evaluate the integral

$$I = \iint_R x e^{-y} dA$$

where R is the region defined by $0 \leq x \leq 2$ and $y \geq 0$.

4. ($1\frac{1}{2}$ pts.) **Calculus:** Consider the system of functions $F = (f_1, f_2)$ where $f_1(x, y) = x^2 + y^2$ and $f_2(x, y) = e^x \sin(xy)$. Compute the Jacobian of F .

5. ($1\frac{1}{2}$ pts.) **Calculus:** Show that the polynomial $P(x) = x^3 + 2x^2 - x - 1$ has 3 roots in $[-3, 1]$.

6. Let $T(x, y, z) = (x + y, y + z)$ be a map from $\mathbb{R}^3$ to $\mathbb{R}^2$.

(a) **(1 pts.)** Is the map linear? Why?

(b) **(3 pts.)** Give a basis for $\ker T$. What is the dimension of $\ker T$?

Hint: $\ker T = \{x \in \mathbb{R}^3 \mid T(x) = 0\}$.

(c) **(1 pts.)** Without finding a basis, which is the dimension of $\operatorname{Im} T$?

(d) **(2 pts.)** Give a basis for $\operatorname{Im} T$.

Hint: $\operatorname{Im} T = \{b \in \mathbb{R}^2 \mid \exists x \in \mathbb{R}^3, Ax = b\}$.

7. Let $B = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$ be a basis for $\mathbb{R}^3$, and let $B' = \{(2, 1, 0), (1, -1, 1), (0, 1, 2)\}$ be another basis for $\mathbb{R}^3$.

(a) **(2 pts.)** Find the change of coordinates matrix ${}_{B \leftarrow B'} P$ that turns coordinates from basis B' to basis B .

(b) **(1 pts.)** Given a vector v with coordinates relative to the basis B' given by $(4, 5, 6)$, find its coordinates relative to the basis B .

(c) **(3 pts.)** Using ${}_{B \leftarrow B'} P$, find the change of coordinates matrix ${}_{B' \leftarrow B} P$ that turns coordinates from basis B to basis B' .

8. **(3 pts.)** Let V be the vector space of all 2×2 square matrices. Let H the subset of V of all the diagonal 2×2 matrices. Show that H is a subspace of V .

9. Let $A = \begin{bmatrix} 7 & 0 & -4 \\ 0 & 5 & 0 \\ 5 & 0 & -2 \end{bmatrix}$.

(a) **(1 pts.)** Find all eigenvalues of A .

(b) **(3 pts.)** Give a basis for every eigenspace of each eigenvalue.

(c) **(1 pts.)** Determine whether A is diagonalizable. If so, give the diagonalization, i.e., an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. If not, explain why A is not diagonalizable.

10. Let $W = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 4 \\ -4 \end{bmatrix} \right\}$ be a subspace of $\mathbb{R}^4$.

(a) **(3 pts.)** Find an orthogonal basis for W .

11. Mark each of the following statements as true or false. If the statement is true, give a proof. If the statement is false, give a proof or provide a counterexample.

(a) **(1 pts.)** Since the set $\{1, t, t^2\}$ has 3 elements and is linearly independent, it is a basis of $\mathbb{P}_3$.

(b) **(1 pts.)** If A and B are two square matrices of the same size, then $2 \det(A + B) = 2 \det A + 2 \det B$.

(c) **(1 pts.)** If u, v are orthogonal vectors in $\mathbb{R}^n$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

